



Car hack warning

With advancement in car tech, FBI and some other organizations have come forward to warn us about them being hacked <http://dgit.in/HackWarn>



Script King

The Stack Overflow survey analysis data proves that JavaScript is still the best the most preferred 1 among non graduate too. <http://dgit.in/ScriptKing>

WATERMARKING

Watermarking, although itself not a full fledged DRM-technology, is a tool used to aid the larger goal of DRM – to identify copyrighted material obtained illegally. It is a form of steganography that embeds metadata or identifying information into media. This embedding is done by altering or adding some information into the underlying code that forms the structure of the digital media.

between files that did and did not violate copyright, encouragement to file sharing by digitizing music in the first place etc) that were struck down one by one accusing Napster of insufficient policing. In September 2001, Napster agreed to settle the case by paying \$26 million and by May 2002, Napster had filed for bankruptcy, thus ending its first incarnation. This set the precedent that any unregulated peer to peer file sharing is illegal and harmful to the original copyright holder.

Hence it hasn't been anything surprising that technology has been repeatedly outpacing the law that governs it in terms of copyright infringement. The MGM Studios vs Grokster case in 2005 is considered by many to be the sequel to the Napster case. Filesharing services like BitTorrent and torrent sites like PirateBay have been facing long legal battles but the opposition has so far been unable to shut these services permanently, thanks to very freedom and anonymity that internet stands for.

All these locks

DRM currently comes in many shapes and sizes. For example, for computer games alone, there have been multiple forms of DRM as shown in the table below

Method	Description	Games
Limited Install activations	Limited installations from a particular disk by verifying each installation over an online server	Mass Effect, Spore, Call of Duty 4, Assassin's Creed, Crysis, Bioshock
Persistent Online Authentication	Constantly downloaded bits of the game from online server to complete otherwise incomplete game	Silent Hunter 5, Assassin's Creed 2, Settlers 7, Diablo 3,
Software Alteration	Game plays out differently if pirated copy is being used	Operation Flashpoint: Cold War Crisis, Serious Sam 3: BFE
Product Keys	A license key is required mid-installation for it to proceed further	Just too many examples!

Another notable area where DRM has a strong-ish hold is eBooks. Adobe, Amazon and Apple that provide the software and the backend infrastructure to support the entire publishing process have their own DRM methods that are designed to restrict copying, alteration and exchange of eBooks from their proprietary devices or applications. In one of the most significant examples in recent times, in July 2009 Amazon remotely deleted copies of George Orwell's Animal Farm and Nineteen Eighty Four from customer's Amazon Kindles and offered them a refund. Amazon cited the reason as copyright violation because the company that was publishing and selling the books on Amazon's platform did not have the rights to do so. They couldn't have chosen a more perfect book right?

The music industry has contributed significantly to the progress of DRM technology. After initial compatibility issues with CDs that came with DRM technology, in 2005 Sony BMG included new DRM technology into their CDs that covertly installed DRM software on the users' computer without explicitly asking for their permission. Along with the obvious issues with this, the software also included a rootkit that introduced a security vulnerability that could be easily exploited by others. Eventually, Sony had to recall most of the CDs that included this software and provide a cash compensation or a free album download to users.

DRM and Music Streaming

With the rise of music streaming in the later 2000's, there came the need for newer forms of DRM. Perhaps the most popular music content provider at the time of its

inception, Apple's iTunes worked with its own FairPlay DRM that allowed music obtained via iTunes to be only played on Apple devices and Apple's Quicktime Player. Apple even launched a higher quality iTunes Plus format in May 2007 that could be obtained at a higher price.

Napster, in its second incarnation also adopted DRM tech. Their method allowed the user to stream an unlimited amount of music while the subscription was valid by downloading the music in WMA format. When the subscription expired, all downloaded music was made unreadable until it was renewed.



Label proposed by the Free Software Foundation for DRM free works

The keys to the locks

One of the most recent testaments to DRM was Steve Jobs' 'Thoughts on Music' open letter back in 2007. In the essay, he cited the requirements of music labels as the only reason why iTunes was still persisting with DRM. In his words -

"Imagine a world where every online store sells DRM-free music encoded in open licensable formats. In such a world, any player can play music purchased from any store, and any store can sell music which is playable on all players. This is clearly the best alternative for consumers, and Apple would embrace it in a heartbeat."

Meanwhile, Apple itself has been on the receiving end of an antitrust lawsuit that accused its iPod devices and iTunes of quashing competition by preventing any other music store from working on their platform.

Even at present, DRM remains one of the most highly debated technologies in the world of content sharing. Either the proponents of intellectual property get to assert their right with the content and impose stricter regulations, or the activists against DRM get to convince the law that the harm which DRM causes to legitimate users is far more than the harm caused by illegitimate users. The true answer to the DRM conundrum shall always be an elusive balance that allows valid cases of reproduction and attribution without robbing the original creator of the benefits that they deserve. 